

Aki Uddin

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Education

University of Michigan — B.S.E. in Mechanical Engineering Ann Arbor, MI
University Honors and Awards; Dean's Honor List (2025, 2026) Aug 2025 – Apr 2029 (Expected)
Cass Technical High School — Diploma, Mechanical Engineering Detroit, MI GPA: 4.3/4.0 2021 – 2025

Technical Skills

CAD & Design: Onshape (part and assembly modeling, .STEP export for manufacturing), SOLIDWORKS; designing to hand-measurement and reach specifications; tolerancing printed parts to off-the-shelf switches and joystick modules

Fabrication: FDM 3D printing (enclosures, keycaps, stick caps, supports); woodworking — cutting, drilling, fastening, adhesives; shop machine use at the Wilson Student Team Project Center

Electronics: Soldering switch and joystick connections, wiring and harness routing inside enclosures, Arduino / Pro Micro firmware in C++, fault isolation on non-responsive inputs

Test & Verification: Force-gauge actuation testing, dimensional verification with calipers and rules, usability test protocol design and results analysis

Engineering Process: Translating user needs into measurable specifications, subject-matter expert interviews, technical report writing

Engineering Projects

Adaptive Game Controller for Hemiparesis Ann Arbor, MI
Product Engineer, Team PlayMakers | University of Michigan, ENGR 100 Jan 2026 – Apr 2026

- Designed a right-hand-primary game controller with a three-person team for a user with hemiparesis whose left hand is significantly weakened, enabling independent play on a standard console title
- Translated three user requirements into verifiable specifications and confirmed each with calipers, rules, and a force gauge: drift and item buttons within **2 cm** of each other, joystick within **9.5 cm** of the cluster, and a left-hand acceleration button measured at **0.96 N** against a 2 N ceiling
- Modeled and 3D printed six custom parts in Onshape — a concave 10 mm stick cap, two 17 mm keycaps toleranced to seat on the switch stems, a 25.5 mm low-precision acceleration button, and three supports — and fabricated a 35 × 25 × 5 cm wooden enclosure with a flat base for tabletop stability
- Ran a **5-participant** usability study at roughly 20 minutes per session; all participants operated every primary function, and feedback on joystick reach and cushioning was captured as prioritized revisions
- Presented the prototype to **two practicing professionals** in adaptive technology and revised the design from their feedback before final assembly

Custom 3D-Printed Game Controller Ann Arbor, MI
Individual Design and Build | University of Michigan, ENGR 100 Winter 2026

- Modified a provided baseline CAD file in Onshape into an individual controller design, dimensioned from a physical controller and scaled to personal hand proportions; added contoured hand grips absent from the baseline to sustain comfort over extended sessions
- 3D printed the full enclosure and button set and delivered final .STEP files for the top case, bottom case, and buttons; soldered, wired, and programmed the interior Arduino circuit independently
- Diagnosed and repaired recurring input failures by isolating cold solder joints and switch misalignment, reworking connections until all inputs passed continuity and function checks

Experience

Math Lab Proctor | University of Michigan, Department of Mathematics Sep 2026 – Present

- Monitor the test-taking process in the Mathematics Department's instructional computer labs, ensuring exam integrity and assisting students with lab logistics

Chassis Sub-team Member | University of Michigan Chem-E-Car Jan 2026 – Present

- Fabricate mechanical components for a shoebox-scale competition vehicle at the Wilson Student Team Project Center and design car parts in CAD, building the housings that carry each chemical sub-team's power and braking system

Involvement & Honors

- Women in Science and Engineering Residence Program (WISE RP)** — 32nd Cohort Member, Jul 2025 – Present; **MASA, 3D Printing Club, Wolverine Athletic Association** — Member, 2026 – Present
- University of Michigan Honors and Awards and Dean's Honor List** — 2025, 2026
- Michigan Seal of Biliteracy**, Michigan Department of Education — Sylheti and Bangla